

Ajitkumar Radder

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Professional Summary

Lead R&D Engineering Professional with 12 years of experience in the full-lifecycle development of off-highway, construction, and electric drivetrains. Specializing in the technical design and integration of complex eAxle and transmission architectures from concept to commercialization. Globally recognized dual-patent holder in electric shifting mechanisms with a proven track record in New Product Development (NPD) and delivering high-performance, cost-effective engineering solutions.

Skills

- ENGINEERING SOFTWARES

Creo & Windchill	Ansys	KISSsoft	Mathcad
Solidworks	CETOL	Siemens NX CAD/CAM	Catia

- TECHNICAL SKILLS

DFMEA	GD&T	2D & 3D stack-up Analysis	Bearing, Seals, Gears
Casting- Forging, Machining	Virtual Validation		

- QUALITY AND COMPLIANCE

ISO 9001, IATF 16949 Audit	ISO, DIN, ANSI standards
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- TECHNICAL LEADERSHIP

Technical Project Execution & Resource Planning	Cross-Functional Engineering Collaboration	Peer Mentorship & Knowledge Transfer	Engineering Documentation & Reports
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Work experience

10/2021 – present
Pune, India

Deputy Manager - R&D (Electrification) Carraro Technologies

- Served as the lead mechanical designer for ongoing axle and transmission development, taking full ownership of the 3D modeling, core calculations, and drawing releases.
- Engineered the new gamma series light-duty construction axles from concept through commercialization. Collaborated with the Luxembourg team to deliver 15 distinct product variants and customized mounting interfaces for specific OEM applications.
- Key member of the engineering team for 160hp electric tractor front axles and 2-speed gearbox attachments, driving the core engineering processes including concept generation, tolerance stack-ups, DFMEA, FEA, and steering analysis.
- Conducted regular design reviews with global headquarters to proactively identify and resolve technical risks early in the development cycle.

International Deputation | Luxembourg (Sep 2025 – Dec 2025):

- Executed the core design and integration of a 40-60HP electric tractor transaxle, working directly with the European engineering team during a 3-month on-site deputation.

02/2018 – 10/2021
Bangalore, India

- Developed mechanical and electro-hydraulic shifting control strategies for both Indian and European markets, resulting in 2 invention disclosures.
- Attended Agritechnica in Germany to evaluate emerging technologies in the off-highway sector and integrate technical learnings into future designs.

Design Engineer II - R&D (Electrification) Meritor Commercial Vehicle Systems Pvt Ltd

- Executed core engineering deliverables across the eAxle design lifecycle from concept to commercialisation, taking care of DFMEA, core calculations, releasing CAD models, tolerance analysis, FEA, and testing support.
- Utilised Six Sigma tools like DoE and Pugh Matrix, along with FEA and Ansys topology optimisation, to improve the designs for shift forks, shifter cams, and planetary carriers.
- Managed the KITAS sensor integration for class 6-7 vehicle drivetrains, covering concept development, tone ring design, stack-up studies for air gap clearance, and coordinating with suppliers.
- Worked on electric axle shifting units and brought in new methodologies for sizing and designing them using tools like Creo Mechanism and Dynacam. The barrel and sector cam designs improved during this time directly led to 2 global patents.
- Executed component and sub-assembly level FEA, and resolved critical tolerance stack-up issues across three different eAxle programs to ensure design integrity, smooth manufacturing, and assembly.
- Coordinated weekly with the US and European design teams to ensure projects stayed aligned and met deadlines.

09/2017 – 02/2018
Bangalore, India

Mechanical Engineer –Material Handling Datamatics (Deputed at Ingersoll Rand)

- Supported design activities like modeling, drawing creation, concept developments, stack up analysis for Winches and Hoists for Ingersoll Rand
- Worked on level-wind mechanism development project for Winches.
- Created product manuals and plant instruction charts to standardise business unit operations.
- Managed Sustenance and engineering change request processing in PLM.

05/2015 – 08/2017
Pune, India

Design Engineer - Aerospace Crane Process Flow Technologies – Pune

- Involved in the design and development of lube and scavenge pumps, fuel flowmeters, and fuel flow transmitters across multiple aerospace programs.
- Executed CAD modeling, drafting, FEA mechanical calculations, and tolerance stack-up analysis for high-profile projects including the Pratt & Whitney AEP, Airbus A320, Kawasaki KHI MGB, GE9X, and GE777X.
- Performed stack load FEA to validate housing strength and Belleville preload specifically for the GE777X project.
- Analysed complex test data and generated comprehensive test reports for the GE9X and Airbus A320 fuel boost pump programs.
- Prepared concept 3D assemblies for RFQs, Bills of Materials (BOM), engineering layouts, and design review documentation to support business unit operations.

08/2014 – 05/2015
Pune, India

Design Engineer Shree Industries

- Designed pressure die casting dies, checking & trimming fixtures, relation gauges using SolidWorks according to specific customer requirements.
- Involved in the design and development of a series of stepper motors utilising reverse engineering and extensive literature surveys.
- Prepared manufacturing drawings for dies and accessories, and attended manufacturability reviews with suppliers and the internal team.
- Developed press tools to trim casting runners and overflows, significantly increasing productivity across the unit.

Trainings and Certifications

12/2020	DFSS Six Sigma Green belt certification Meritor Inc
06/2019	Ansys Workbench Software training Ansys
12/2020	Project Management Basic Training Meritor Inc
01/2024	GD&T and Tolerance Stack-Up analysis Symphony Tech Pune
03/2025	eMobility KISSsys Drivetrain Modeling Workshop FEDERTEC Academy

Education

06/2010 – 07/2014 Bangalore, India	B.E. Mechanical Engineering - (7.63GPA) R. V. College of Engineering Mechanical Lead - Chimera Racing During Engineering, I served as Mechanical Lead for student formula hybrid racing team. Leading chassis design, suspension, and drivetrain systems, contributing to key achievements in high-performance competitions.
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Patents

02/2021	Axle assembly having a barrel cam USPTO, EU, BR, CN US11371588B1, EP4047247A1, BR102022003227A2, CN114962577A This patent relates to an e-shifter mechanism with a barrel cam, specifically engineered for application in Meritor's I7Xe eAxle AMT Transmission.
02/2021	Axle assembly having a sector cam USPTO, EU, BR, CN US11608877B2, EP4047248A1, BR102022003234A2, CN114962563A This patent relates to an e-shifter mechanism with a sector cam, specifically engineered for application in Meritor's I4Xe eAxle AMT planetary Transmission

Publications

01/2015	Parametric study of pulsed CO2 Laser surface modification of alumina ceramics Journal of Manufacturing science and production Researched the impact of laser parameters (power, frequency, speed) on material removal & surface quality in CO2 laser treatment of alumina ceramics. Employed ANOVA, S/N ratio, Response surface methodology (RSM), & Grey relational analysis (GRA). Published findings to advance ceramic processing.
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